Registration

Indian Institute of Technology Bhubaneswar
Autumn End-Semester Examination 2018

Subject: EC2L002-Signals and Systems
Date: 30-11-2018

Full Marks: 100
Duration: 03 Hours

Answers All Questions (Assume parameters if required and Use properties of transform)

PART-I (30 × 1 =30 Marks)

1. The examples for harmonic signals are_____, and _____.
2. The even part of a function $x[n] = u[n] + u[-n]$ is_____.
3. The signal operations performed on the independent variable are_____, _____ and _____.
4. The mathematical expressions for continuous-time real exponential decaying and exponentially damped sinusoidal signals_____ and _____.
5. The relationship between the signum and unit step functions is_____.
6. The relationship between the impulse responses of a system and its inversion system is_____.
7. The Laplace transfer functions ideal integrator and ideal differentiator are _____ and _____, respectively.
8. The initial and final values of $x(t)$ are _____ and _____ if $X(s) = e^{-s}\frac{-2}{s(s+2)}$.
9. The $X(\omega)$ of $x[n]$ can be computed if ROC of $X(z)$ contains_____. The relationship between the Z transform and DTFT is _____.
10. The frequency range of audio signal is_____ and pitch frequency range of human voice is_____.
11. Sampling is the process of _____ and quantization is the process of_____.
12. The difference equation of echo filter_____ and its impulse response_____.
13. The discrete-time aperiodic has _____(continuous/discrete) and _____(periodic/aperiodic) spectrum.
14. The frequency of a powerline signal is _____ (Hz) and its autocorrelation has a period of _____(second)
15. _____ (IIR/FIR) system is always stable and its Mth order difference equation is _____.
16. The example for an incrementally linear system is___,which is an example for___(invertible/non-invertible) system.
17. The modulation property of the CTFT is define as_____ and the correlation property of DTFT is defined as _____.
18. _____ filter can provide flat response in passband and _____ filter can provide maximally linear phase response.
19. Consider the analog signal $x(t) = 20\sin(100\pi t) + 5\cos(3000\pi t) - 5\sin(5000\pi t)$. The Nyquist rate for this signal is_____ and its discrete version, $x[n]$ is_____.
20. The zero-state response is defined as_____ and zero-input response is defined as_____.
21. For an input signal $x(t) = \sin\left(500t + \frac{\pi}{3}\right)$, the output of a CT system with an impulse response $h(t) = \delta(t)$ is_____.
22. A zero exists at z = _____ if $X(\infty) = 0$ and a pole exists at z = _____ if $X(\infty) = \infty$.
23. In transfer function of discrete-time system, if the order of the denominator polynomial (N) is less than the numerator polynomial (M) then the _____ number of _____(zeros or poles) occur at _____(origin / infinity).
24. If a single positive real-valued pole is located within an unit circle then a signal is ___(real/complex) and exponentially _____(decaying/rising) in time-domain.
25. In the Z-plane, a pair of complex-conjugate poles located on the unit circle results in _____(sinusoidal/exponentially decaying sinusoidal/ exponentially rising sinusoidal) signal.
26. The examples for the orthogonal functions are _____ and _____.
27. Assuming that a 3-bit ADC channel accepts analog input ranging from 0 to 5 volts, the number of quantization levels is_____ and step size of the quantizer is_____.
28. The Fourier series coefficients of $x(t) = 10\sin\left(100\pi t + \frac{\pi}{4}\right)$ are_____.
29. The frequency of the 100th Fourier transform coefficients is _____when a sampling rate is 1000 Hz and number of samples is 256 and its frequency resolution is _____.
30. Assume that 1 second voiced signal processed with a frame duration of 30 ms and frame shifting of 10 ms. A total number of frames is _____ and a number of samples per frame is _____ if a sampling rate 16000 Hz.

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PART-II (10 × 2 =20 Marks)

31. Find the fundamental period of the signal $x(t) = 20\sin(100t) + 5\cos(300t) - 5\sin\left(500t + \frac{\pi}{3}\right)$
32. Write the four properties of impulse function.
33. Suppose that the cumulative average of a signal $x(n)$ is computed as $y(n) = \frac{1}{n+1}\sum_{k=0}^{n} x(k)$, $n = 0,1,2 \ldots$. For $n = 5$, the recursive system requires ____ multiplication(s), ____addition(s)_____memory location(s) whereas the non-recursive system requires ____ multiplication(s), ____addition(s)_____memory location(s).
34. Write eight properties of CTFT with suitable mathematical expressions.
35. Sketch the ROC of the Laplace transform of $x(t) = e^{-b|t|}$.
36. Write four properties of ROC of the Z transform.
37. Find the Z transform and ROC of the signal $x[n] = \begin{cases} a^n, & 0 \le n \le N-1, \quad a > 0 \\ 0, & Otherwise \end{cases}$
38. Draw ideal magnitude responses of four types of filters and with a proper labelling of design parameters.
39. What are advantages of the digital signal processing over analog signal processing?
40. Derive the transfer function of a first order RC low-pass filter.

$\frac{1}{s+a}$ $e^{-at}u(t)$ $v(t) = \frac{1}{L}\left(u(t)\right) + v(0^-)u(t)$

$v(t) =$

PART-III (50 Marks)

41. By using Laplace (s-domain) equivalent analysis, **(A)** calculate the voltage $v(t)$ shown in Fig. 1, given an initial current $i(0^{-1}) = 1\,A$. **(B)** Sketch the poles and zeros of the Laplace transform function. **[10 Marks]**

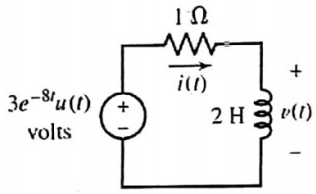

Fig.1:R-L Network

42. Consider an LTIC system with impulse response given by $h(t) = \frac{10}{\pi}\, sinc\left(\frac{10t}{\pi}\right)$. Sketch the impulse response and its transfer function $H(j\Omega)$. Determine the output response of the system for the following inputs: **[10 Marks]**

    (A) $x_1(t) = 2\sin(5t)$; **(B)** $x_3(t) = \sin(8t) + \sin(20t)$

43. Using the Z-transform, determine the response of the system $y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n]$ to the input signal $x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$. **[10 Marks]**

44. Draw a second-order unity-gain Sallen-Key low-pass filter. Assume equal resistor values and equal capacitor values, derive an expression for general transfer function. Design a second-order unity-gain Butterworth low-pass filter with a corner frequency of $f_c = 3\,kHz$. **[10 Marks]**

45. **(A)** By using the application of discrete Fourier transform, explain the algorithmic steps with suitable mathematical expressions for a removal of specific frequency range from the input signal. **[5 Marks]**

    (B) Based on the course topics which were discussed in the classroom, describe a method for finding the pitch period of a voiced signal. **[3 Marks]**

    (C) Suggest FIR filters which can be used for smoothing and emphasizing high-slope portions in a signal. **[2 Marks]**

$a^n u(n)$ $\frac{1}{1-az^{-1}}$ $\sum x(n)e^{-z}$ $\sum x(n)e$

$x(0)z^0 + az^{-1} + \ldots$ $x(1) = a$

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$e^{-a}u(t)$

$\frac{1}{1-x(1)z^{-1}} = \frac{z}{z-1}$ $\frac{6z^2+5z+1}{6}$ $\left(z+\frac{1}{3}\right)\left(z+\frac{1}{2}\right)$ $1 - \frac{1}{3}z^{-1}$ $\frac{(3z+1)(2z+1)}{6}$

$\frac{-5 \pm \sqrt{25-4(6)}}{2(6)}$

$= \frac{-5 \pm \sqrt{1}}{12}$

$\frac{z}{z-a}$ $\sum_0^\infty e^{an}z^{-n}$ $\frac{ar^{n-1}}{(1-r)}$ a^{-n} a^{-n}

$1 + e^{a}z^{-1} + \ldots$ $\frac{a(1-r^n)}{(1-r)}$

$\cos\theta + i\sin\theta = e^{i\theta}$

$\cos\theta - i\sin\theta = e^{-i\theta}$

$2i\sin\theta = e^{i\theta} - e^{-i\theta}$

$a\left(\frac{1-r^n}{r-1}\right)$

$\frac{1}{r-1}$ $= \frac{-5+1}{12}$ $\frac{-6}{12}$ $= \frac{-1}{3}, \frac{-1}{2}$

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